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EXP10:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

/* Playfair matrix given in the question (I/J share a cell) */
char mat[5][5] = {
    {'M', 'F', 'H', 'I', 'K'},
    {'U', 'N', 'O', 'P', 'Q'},
    {'Z', 'V', 'W', 'X', 'Y'},
    {'E', 'L', 'A', 'R', 'G'},
    {'D', 'S', 'T', 'B', 'C'}
};

void find(char ch, int *r, int *c) {
    int i, j;
    if (ch == 'J') ch = 'I';
    for (i = 0; i < 5; i++)
        for (j = 0; j < 5; j++)
            if (mat[i][j] == ch) { *r = i; *c = j; return; }
}

int main() {
    char line[200], pt[400], dg[400], ct[400];
    int n = 0, m = 0, i, j;

    printf("Playfair matrix:\n");
    for (i = 0; i < 5; i++) {
```

```
    for (j = 0; j < 5; j++) printf("%c ", mat[i][j]);
    printf("\n");
}
printf("\nEnter message: ");
fgets(line, sizeof line, stdin);
for (i = 0; line[i]; i++) if (isalpha(line[i])) pt[n++] = toupper(line[i]);
pt[n] = '\0';

/* form digraphs: insert X between double letters, pad last with X */
for (i = 0; i < n; i++) {
    dg[m++] = pt[i];
    if (i + 1 < n && pt[i] == pt[i + 1]) dg[m++] = 'X';
}
if (m % 2) dg[m++] = 'X';
dg[m] = '\0';

printf("Digraphs   : ");
for (i = 0; i < m; i += 2) printf("%c%c ", dg[i], dg[i + 1]);
printf("\n");

for (i = 0; i < m; i += 2) {
    int r1, c1, r2, c2;
    find(dg[i], &r1, &c1);
    find(dg[i + 1], &r2, &c2);
    if (r1 == r2)      { ct[i] = mat[r1][(c1 + 1) % 5]; ct[i + 1] = mat[r2][(c2 + 1) % 5]; }
    else if (c1 == c2) { ct[i] = mat[(r1 + 1) % 5][c1]; ct[i + 1] = mat[(r2 + 1) % 5][c2]; }
    else               { ct[i] = mat[r1][c2];           ct[i + 1] = mat[r2][c1]; }
}
ct[m] = '\0';
printf("Ciphertext : ");
```

```
    for (i = 0; i < m; i += 2) printf("%c%c ", ct[i], ct[i + 1]);
    printf("\n");
    return 0;
}
```

OUTPUT:

```
Playfair matrix:
M F H I K
U N O P Q
Z V W X Y
E L A R G
D S T B C

Enter message: Must see you over Cadogan West. Coming at once.
Digraphs   : MU ST SE XE YO UO VE RC AD OG AN WE ST CO MI NG AT ON CE
Ciphertext : UZ TB DL ZR WQ NP ZL GB ET QA LO ZA TB TQ FK QL TH PO DG
```